

HIMKAVACH — HARDWARE ARCHITECTURE & VALIDATION BLUEPRINT

PS SIH26049 (DRDO) · Team ALPHA 20 · Team ID 131984 · Theme: Smart Automation · Category: Hardware

1 Document control

Document ID	HIMKAVACH-BLUEPRINT Rev B (V9 — red-team corrections; supersedes Rev A)
Date	30 September 2026
Status	Draft for internal review — a validation-pathway blueprint, not a test report
Classification	Student engineering document. Not DRDO-certified. No DRDO logo used.
Computation source	simulation/ physics library (github.com/jaisogani-ai/SIH2026), re-run 30 Sep 2026 — see computation log. Note: the deployed app is a separate implementation with its own scenario parameters (see §5).
Live prototype	https://himkavach.grok.me (simulation only)

2 Scope — what this document is

- A defence-style engineering blueprint for the HIMKAVACH hardware programme: master system architecture, proposed test PCB, physical validation rig, data-provenance architecture, failure/mitigation matrix, traceability, BOM, risk register and phased roadmap.
- It does NOT report measurements. No PCB has been fabricated, no components purchased, no sensor data recorded. Every value below carries exactly one evidence tag.
- Simulator output is labelled IMPLEMENTED DIGITAL PROTOTYPE. Current MVP: simulation/design screening; physical validation planned.

3 Evidence legend — the only six classes used

Tag	Meaning	Where it appears in this document
MEASURED	Physical sensor data from built hardware	Nowhere yet — the track is specified, not populated
COMPUTED	Produced by running the team's code	Atmosphere, Paschen, thermal, battery outputs (re-run 30 Sep 2026)
LITERATURE	Published standard, paper or datasheet value	IEC 60664-1 (secondary), IPC-2221B, US Std Atmosphere 1976, PIB releases
ASSUMPTION	Team modelling choice, flagged for calibration	Convection exponent 0.8, risk-index thresholds
ESTIMATE	Unverified planning value	All prices — VERIFY BEFORE PURCHASE
PROPOSED / FUTURE VALIDATION	Designed but not yet built / run	Test PCB, rig, chamber, E1/E2/E3 experiments

Taxonomy note: IMPLEMENTED and SIMULATED appear in this document only as plain status words (e.g. "IMPLEMENTED DIGITAL PROTOTYPE"), never as bracketed evidence tags. Every bracketed tag is one of the six classes above.

NO PHYSICAL MEASUREMENTS EXIST

No PCB fabricated · no components purchased · no sensor data recorded · no laboratory evidence fabricated

4 Programme status (README)

HIMKAVACH-BLUEPRINT Rev B (V9) · 30 Sep 2026 · Student engineering document. Not DRDO-certified. No DRDO logo used. Simulation output, no physical hardware validation has yet been incorporated.

• **IMPLEMENTED:** web simulator with physics engine, risk index and comparison tables (screenshot §5); all outputs [COMPUTED] and badged as such.

• **PROPOSED:** TEST-PCB-01 layout, validation rig, E1 (thermal), E2 (HV, simulation-only), E3 (cold battery).

MASTER SYSTEM ARCHITECTURE — FIVE LAYERS

Environment → digital physics → test PCB → physical validation → data provenance

HIMKAVACH — MASTER SYSTEM ARCHITECTURE

PS SIH26049 (DRDO) · Team ALPHA 20 · 5-layer reliability screening & validation system

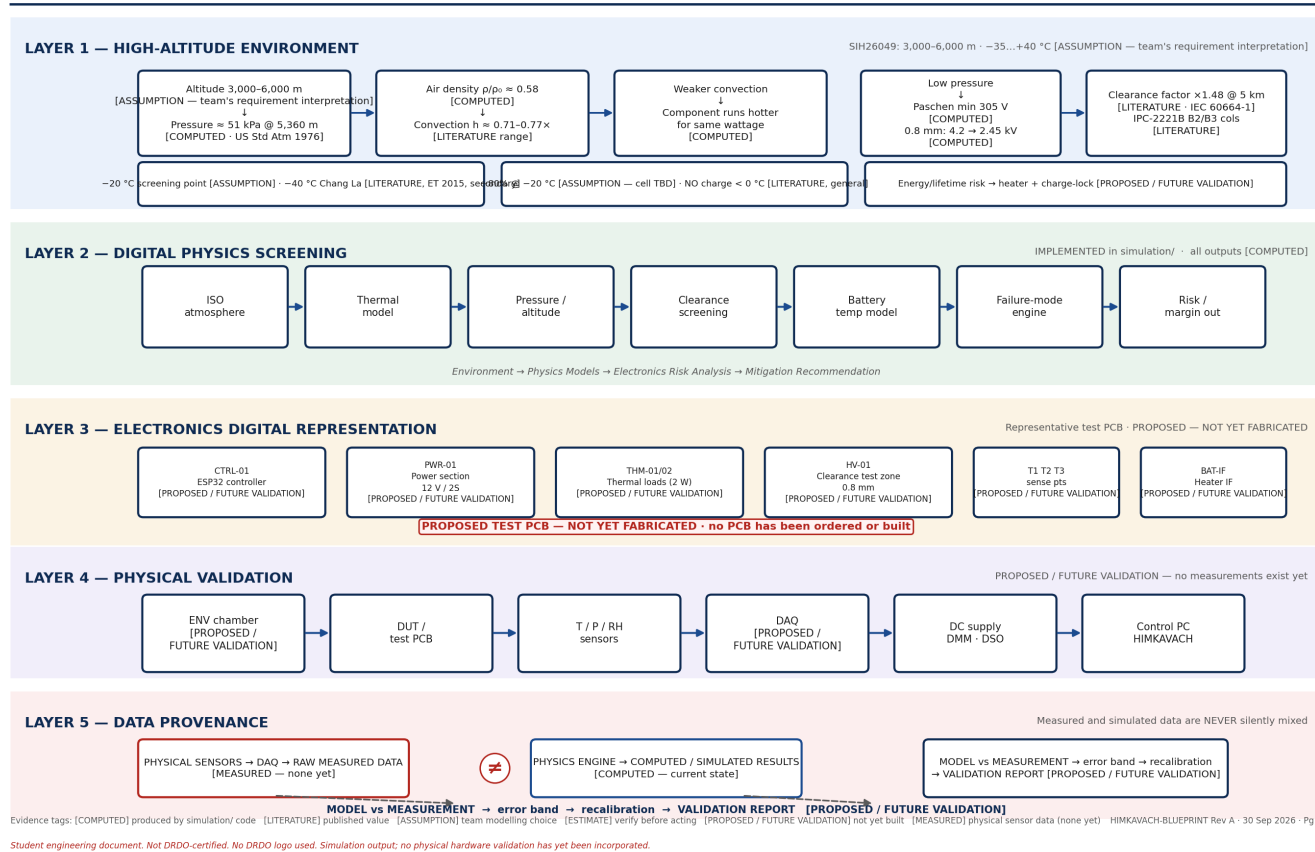
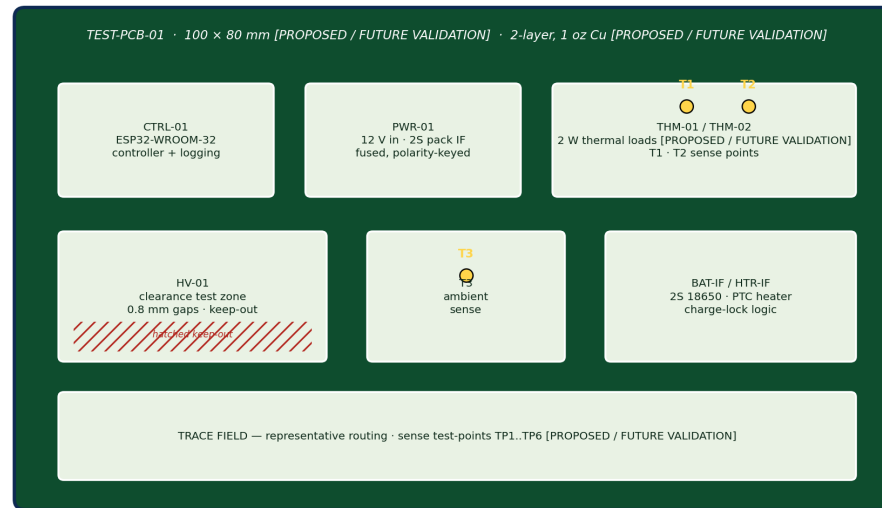


Figure 2.1 — five-layer system architecture. Evidence tags on every value; measured and simulated tracks never merge.

PROPOSED TEST PCB — LAYOUT & PLACEMENT

TEST-PCB-01 · PROPOSED TEST PCB — NOT YET FABRICATED · no PCB ordered or built

PROPOSED TEST PCB — TOP VIEW (representative, not to scale)



PROPOSED TEST PCB — NOT YET FABRICATED · no PCB ordered · no measurements exist

Purpose: give E1/E3 a repeatable physical specimen whose geometry matches the clearance & thermal models.

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Placement rationale (all zones PROPOSED)

- **CTRL-01 (ESP32-WROOM-32)**: controller + logging; antenna keep-out observed; 3.3 V logic throughout.
- **PWR-01**: 12 V in and 2S pack interface, fused and polarity-keyed; heater return kept off the logic ground star point.
- **THM-01 / THM-02**: 2 W thermal loads placed symmetrically; T1/T2 sense points resolve component vs local-ambient temperature for E1 calibration.
- **HV-01**: clearance test zone with 0.8 mm gaps, hatched keep-out — no copper pour, no components; geometry matches the Paschen screening model (§7). E2 stays simulation-only; no arcing is ever run on a student rig.
- **T3**: ambient sense placed away from thermal loads.
- **BAT-IF / HTR-IF**: 2S 18650 interface, PTC heater drive and charge-lock logic for E3 cold-battery work. Pack: 2S1P, 7.4 V nominal / 8.4 V max / 6.0 V cutoff, 22.2 Wh (3000 mAh-class cells — ASSUMPTION, cell TBD before purchase).
- **Trace field**: representative routing with sense test-points TP1..TP6 for DMM/DSO probing.

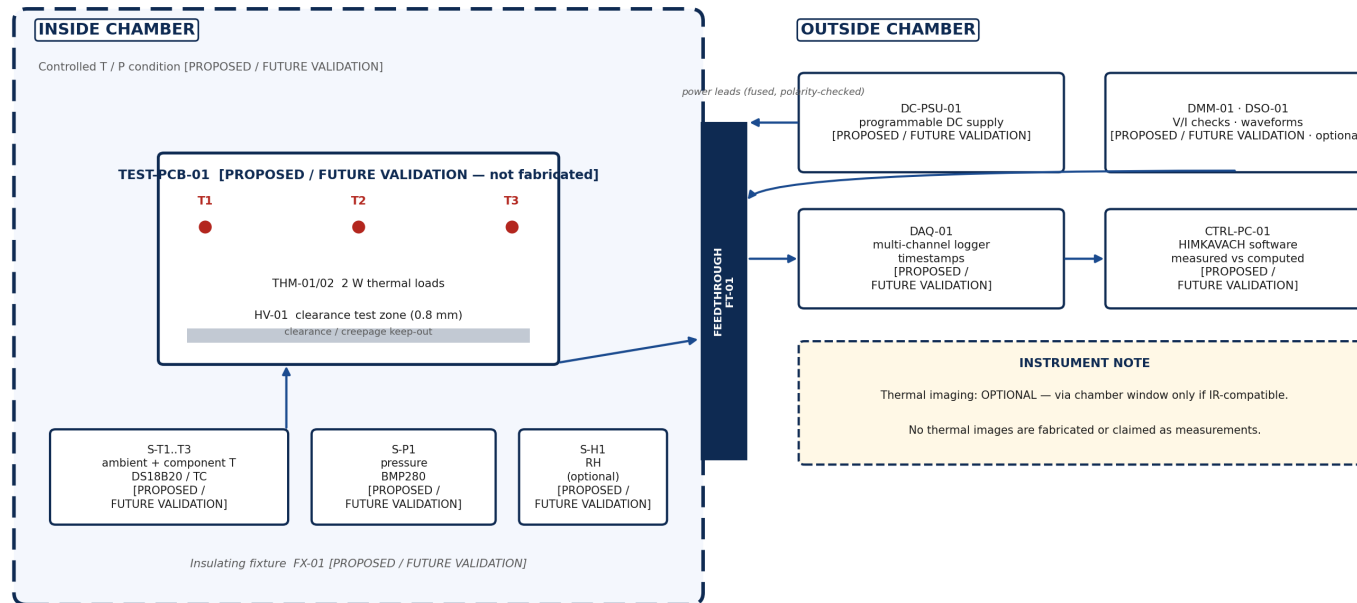
Drawing is representative and not to scale. Fabrication starts only after the Phase-1 design review (§9).

PHYSICAL VALIDATION TEST RIG

PROPOSED / FUTURE VALIDATION — no hardware built, no measurements taken

PHYSICAL VALIDATION TEST RIG

PROPOSED / FUTURE VALIDATION — no hardware built, no measurements taken



- ① Chamber creates the T/P condition — E2 HV arcing never runs in a student rig.
- ② T1/T2/T3 resolve ambient vs component temperature for E1 calibration.
- ③ S-P1 logs pressure; hand-pump jar reaches ≈17 kPa abs [COMPUTED] — room-temp only, leak-down.
- ④ FT-01 feedthrough: power/signal cross the wall; chamber never opened mid-run.
- ⑤ DAQ timestamps every channel; raw file is immutable — calibration is a separate layer.
- ⑥ CTRL-PC-01 keeps MEASURED and COMPUTED datasets in separate stores.

Evidence tags: [COMPUTED] produced by simulation/ code [LITERATURE] published value [ASSUMPTION] team modelling choice [ESTIMATE] verify before acting [PROPOSED / FUTURE VALIDATION] not yet built [MEASURED] physical sensor data (none yet) HIMKAVACH-BLUEPRINT Rev A · 30 Sep 2026 · Pg 4

Student engineering document. Not DRDO-certified. No DRDO logo used. Simulation output; no physical hardware validation has yet been incorporated.

Figure 4.1 — inside vs outside the chamber. E2 HV arcing never runs on a student rig.

DIGITAL PROTOTYPE — SOFTWARE VALIDATION LAYER

IMPLEMENTED DIGITAL PROTOTYPE · Current MVP: simulation/design screening; physical validation planned

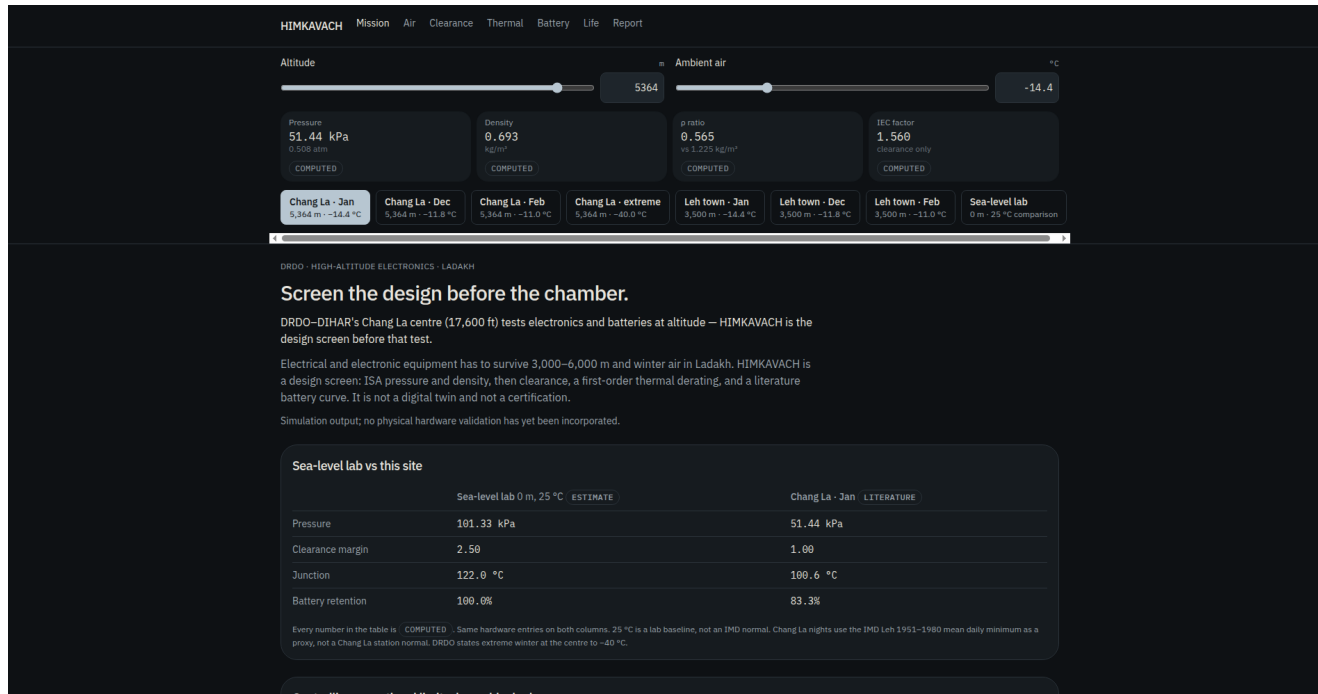


Figure 5.1 — actual screenshot of <https://himkavach.grok.me> (Mission page), captured 30 Sep 2026. Not generated, not reconstructed.

Software data flow (summary — full architecture §6)

Inputs: LITERATURE parameters (standards) + ASSUMPTION values (model choices) → physics engine → COMPUTED results → figures & risk index → versioned computed dataset. No measured data enters this path; when the rig runs, measured data lives in a separate store and the two are compared, never merged.

What is implemented

- Reference physics library (simulation/common/physics.py): standard atmosphere, Paschen clearance, lumped thermal model, linear battery derating, risk index. The deployed app is a separate implementation with its own scenario parameters — see “Reading the screenshot”.
- Web simulator: altitude/ambient sliders, mission presets (Chang La, Leh, sea-level lab), computed stat cards, sea-level comparison table.
- Every output on the site is badged [COMPUTED]; the site footer states: “Simulation output; no physical hardware validation has yet been incorporated.”
- Graphs in repo (screenshots/graphs/): Paschen curve, clearance analysis, pressure/temperature vs altitude, thermal convection penalty, battery derating — all [COMPUTED] simulation figures.

Reading the screenshot

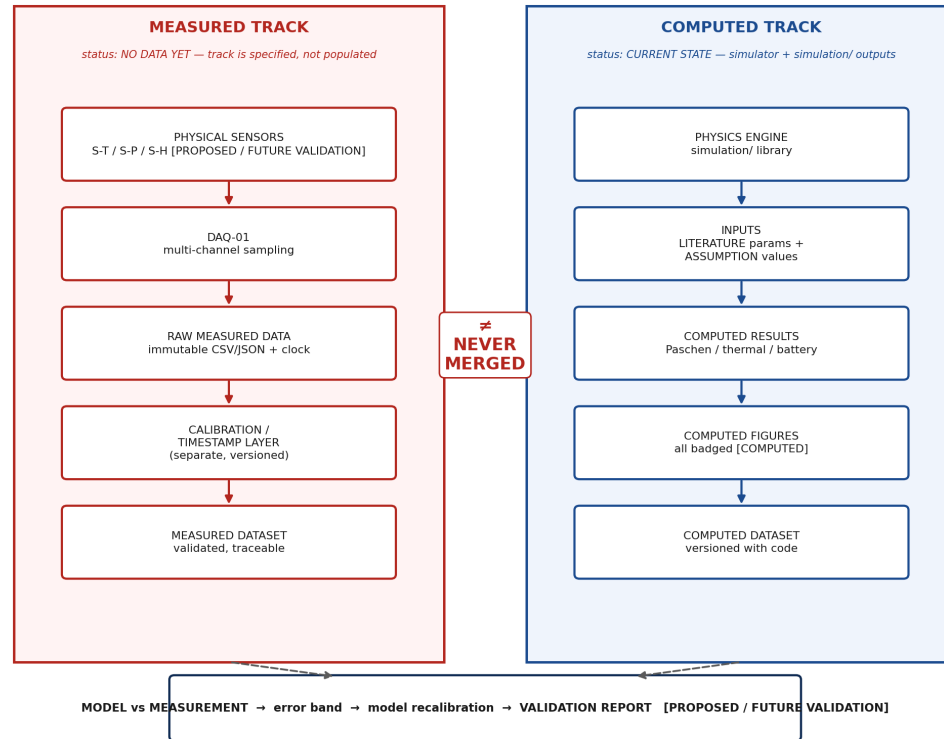
- Stat cards: $P = 51.44$ kPa, density 0.693 kg/m³, p ratio 0.565 , IEC factor 1.560 at $5,364$ m — all [COMPUTED]. Site scenario: 100 Wh placeholder pack [ASSUMPTION] (not the 2S1P 22.2 Wh hardware), 20 W load, Theta-JA thermal (8 W, 12 °C/W, $\times 1.20$ at $5,364$ m), clearance 1.5 mm / 100 V.
- Two implementations, both [COMPUTED], never mixed: the blueprint's §7.2 anchors come from simulation/ (ISA; linear battery 80% @ -20 °C; lumped 2 W thermal). The screenshot's battery/thermal readouts come from the app (lookup-table 77.4% @ -20 °C; Theta-JA junction). Comparison table (sea-level lab vs Chang La): all [COMPUTED], not measured; the clearance-margin formula is app-source-only — documenting it is a standing action (R-09).
- Caption correction: the site attributes electronics testing to “DRDO-DIHAR's Chang La centre” — DIHAR Leh is an agro-animal / life-sciences establishment, not an electronics-hardening lab (2015 ET, secondary — see §9). Screenshot reproduced verbatim.

HARDWARE → SOFTWARE DATA ARCHITECTURE & PROVENANCE

Measured and simulated data are never silently mixed

HARDWARE → SOFTWARE DATA ARCHITECTURE

Provenance: measured data and simulated data are never silently mixed



Doctrine: the model loses every tie. Discrepancy → report, never silent overwrite.

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Provenance rules

- Two tracks, two stores. MEASURED track: sensors → DAQ-01 → immutable raw file (CSV/JSON + clock). COMPUTED track: physics engine → versioned outputs. A comparison reads both; nothing writes across.
- Raw measured data is immutable. Calibration, timestamps and unit conversions live in a separate, versioned layer — calibration never edits raw.
- Computed datasets are versioned with the code commit that produced them (computation log, re-run 30 Sep 2026).
- Doctrine: the model loses every tie. A discrepancy between measurement and model is reported as an error band and used to recalibrate — never silently overwritten.
- Status today: the MEASURED track is specified end-to-end and empty. The COMPUTED track is the current state of the programme.

Why this matters

A DRDO reviewer reading a future validation report must be able to tell, for any number, whether it was measured on the rig or computed by the simulator, which code version produced it, and which calibration was applied. This architecture makes that answer mechanical, not a matter of trust.

FAILURE MECHANISMS, MITIGATIONS & VALIDATION

Every mechanism names its validation experiment

7.1 Failure-mechanism → mitigation → validation matrix

ID	Failure mechanism	Mitigation (design)	Validation
FM-01	Thin air → weaker convection → higher junction ΔT for same wattage [COMPUTED: 18.7 K rise at sea level → 29.1 K at 5,360 m, 2 W load]	Thermal model + copper pour + power derating; convection exponent 0.8 flagged [ASSUMPTION]	E1 thermal experiment [PROPOSED / FUTURE VALIDATION]; calibrates exponent
FM-02	Low pressure → lower Paschen breakdown: 0.8 mm gap 4.198 kV → 2.446 kV [COMPUTED]; clearance factor $\times 1.48$ @ 5 km [LITERATURE, IEC 60664-1, secondary]	IPC-2221B B2/B3 altitude columns [LITERATURE]; HV-01 keep-out zone on TEST-PCB-01 [PROPOSED / FUTURE VALIDATION]	E2 — simulation-only. No HV arcing on a student rig (safety rule)
FM-03	Subzero → battery capacity loss (0.80 usable @ -20°C [COMPUTED from repo model — anchor ASSUMPTION until a specific 18650 datasheet is selected]); charging below 0°C plates lithium [LITERATURE, general manufacturer guidance]	PTC heater + charge-lock logic (BAT-IF/HTR-IF) [PROPOSED / FUTURE VALIDATION]	E3 cold-discharge experiment [PROPOSED / FUTURE VALIDATION]
FM-04	Condensation / icing during thermal cycling → leakage, corrosion [LITERATURE, general]	Conformal coating [PROPOSED / FUTURE VALIDATION]; drainage/shelter in fixture design	Chamber thermal cycling [PROPOSED / FUTURE VALIDATION]
FM-05	Sensor drift at temperature extremes → false margins [LITERATURE, datasheets]	Calibration layer separate from raw data; per-run reference checks [PROPOSED / FUTURE VALIDATION]	E1/E3 calibration runs [PROPOSED / FUTURE VALIDATION]

7.2 Key computed anchors (re-run 30 Sep 2026 — computation log on file)

- 51.47 kPa, $p/p_\square = 0.578$ at 5,360 m [COMPUTED, U.S. Standard Atmosphere 1976 troposphere].
- Paschen minimum from code constants ($A=15$, $B=365$, $\gamma=0.01$): 305.3 V [COMPUTED] vs ≈ 327 V commonly quoted in textbooks; the exact minimum depends on the (A , B , γ) set — ours gives 305.3 V, and the deployed app agrees (same B/A ratio) [COMPUTED]. Screening uses the code's own curve consistently.
- $V_{bd}(0.8\text{ mm})$: 4.198 kV (sea level) → 2.446 kV (5,360 m) [COMPUTED].
- Convection: team model $p^{0.8}$ gives $0.645\times$ at 5,360 m [ASSUMPTION + COMPUTED]; textbook natural-convection scaling $p^{0.5}\dots p^{(2/3)}$ gives $0.694\text{--}0.760\times$ [LITERATURE] — the model is deliberately conservative and E1 exists to calibrate it.
- Battery usable fraction (repo model): 1.00 @ $\geq 0^\circ\text{C}$, 0.80 @ -20°C , 0.25 @ -40°C [COMPUTED from ASSUMPTION-anchored model — no 18650 datasheet on file; cell TBD]. The deployed app uses its own lookup-table curve (77.4% @ -20°C) — see §5; the two value sets are never mixed.
- Pack (single architecture, §8): 2S1P 18650, 7.4 V nominal / 8.4 V max / 6.0 V cutoff, 22.2 Wh rated (3000 mAh-class cells [ASSUMPTION]); 2S BMS + TP5100 2S charger; charge-lock below 0°C [PROPOSED].
- Reference conditions: all anchors above are computed at ISA reference conditions (5,360 m: 51.47 kPa, -19.84°C) unless stated. The live site lets ambient deviate from ISA (§5 screenshot: -14.4°C ambient).
- E1 heat source: the BOM's 12 V ~ 30 W PTC heaters are run derated (PWM via IRLZ44N) to deliver the 2 W test dissipation used in the model — the 30 W rating is heater capacity, not the test load [ASSUMPTION].
- Hand-pump vacuum reference: 63.5 cm Hg vacuum → $\approx 16.6\text{--}17$ kPa absolute [COMPUTED conversion of a catalogue spec — not a measured capability of the proposed rig; achievable pressure depends on pump, seals, leakage, gauge and altitude].

REQUIREMENT TRACEABILITY & BILL OF MATERIALS

Every requirement traces to an artefact · Status: PLAN — nothing purchased · all prices ESTIMATE — VERIFY BEFORE PURCHASE

8.1 Requirement traceability (PS SIH26049)

Bands 3,000–6,000 m and –35...+40 °C are the team’s requirement interpretation [ASSUMPTION]; the PS states HAA/SHAA subzero temperature and low pressure without numeric bands.

Req	Requirement (problem statement)	Simulator (IMPLEMENTED DIGITAL PROTOTYPE)	Test PCB [PROPOSED / FUTURE VALIDATION]	Experiment [PROPOSED / FUTURE VALIDATION]	Standard [LITERATURE]
R1	Operate 3,000–6,000 m	Atmosphere + IEC factor modules	—	E1/E2	IEC 60664-1 Table A.2 (secondary); IPC-2221B
R2	Subzero operation (–35...+40 °C band)	Thermal + battery-temp models	THM-01/02, T1–T3	E1, E3	MIL-STD-810H 502.7
R3	Reliability / lifespan screening	Failure-mode engine + risk index	—	E1–E3 calibration loop	MIL-STD-810H (general)
R4	Low-pressure electrical integrity	Paschen / clearance screening	HV-01 keep-out (0.8 mm)	E2 sim-only	IEC 60664-1; IPC-2221B B2/B3
R5	Battery energy & lifetime at cold	Battery derating curve	BAT-IF / HTR-IF	E3	Cell datasheet — TBD [ASSUMPTION]

Battery architecture — one consistent definition

2S1P 18650 · 7.4 V nominal / 8.4 V max / 6.0 V cutoff · 22.2 Wh (3000 mAh-class) · 2S BMS · TP5100 2S charger · charge-lock < 0 °C [PROPOSED]. Cell TBD — ASSUMPTION — VERIFY BEFORE PURCHASE.

8.2 Hardware bill of materials — realistic prototype plan

ID	Component	Specification	Purpose	Qty	Est. ■	Status
CTRL-01	ESP32 dev board	ESP32-WROOM-32	Controller + logging	2	1,300	Not purchased
THM-01/02	PTC heater	12 V PTC, ~30 W max, run derated to 2 W test load via PWM [ASSUMPTION]	Controlled heat source, E1	2	500	Not purchased
THM-DRV	MOSFET module	IRLZ44N logic-level (see §9.1 N1)	Heater switching from 3.3 V GPIO	2	280	Not purchased
BAT-01	18650 cells	3.7 V 3000 mAh, branded, with datasheet (see §9.1 N3) — ASSUMPTION: CELL TO BE VERIFIED BEFORE PURCHASE	2S1P pack, E3	2 + 2 spares	1,400	Not purchased
BAT-02	BMS	2S 18650 protection board (HX-2S class)	Pack protection	1	180	Not purchased
BAT-03	Charger module	TP5100 set to 2S / 8.4 V mode (see §9.1 N2)	2S pack charging	1	150	Not purchased
S-T1..T3	Temp sensors	DS18B20, waterproof	Thermal array, E1	4	600	Not purchased
S-P1	Pressure sensor	BMP280 module	Ambient pressure logging	1	250	Not purchased
S-VI	Power monitor	INA219 module	V/I logging (E1, E3)	2	440	Not purchased
UI-01	Display	0.96" OLED, I2C	Local readout	1	280	Not purchased
LOG-01	Storage	microSD module + card	CSV data logging	1	350	Not purchased
PSU-01	Power supply	12 V 5 A adapter	Bench power	1	450	Not purchased
MISC	Wiring & hardware	Wires, perfboard, enclosure, fixture	Assembly	lot	600	Not purchased

COMPONENT RATIONALE & RISK REGISTER

Why each choice survived review · what can go wrong and its disposition

9.1 Selection rationale & red-team corrections

- **N1 — MOSFET:** the earlier draft listed “IRFZ44N logic-level”. IRFZ44N is NOT logic-level (needs ~10 V gate drive) and cannot be driven from ESP32 3.3 V GPIO. Corrected to IRLZ44N (logic-level, Vgs(th) 1–2 V) [LITERATURE, datasheets — attach manufacturer datasheet at purchase].
- **N2 — charger:** the earlier draft listed a TP4056 for a 2S pack. TP4056 is a single-cell (4.2 V) charger and cannot charge a 2S pack. Corrected to TP5100 configured for 2S (8.4 V) plus the 2S BMS for protection [LITERATURE, IC datasheets — attach manufacturer datasheet at purchase].
- **N3 — cells:** branded 18650 with a published datasheet only; unbranded cells are excluded from battery tests as a safety rule. Largest cost driver — do not economise here. Exact cell TBD — ASSUMPTION — CELL TO BE VERIFIED BEFORE PURCHASE; no cell datasheet is on file.
- **N4 — E2 needs no HV hardware:** it is simulation-only by student-safety rule. No HV supply appears in this BOM deliberately.
- **N5 — chamber time** (accredited environmental chamber for combined low-pressure + temperature per JSS 55555 Test No. 3) is far outside the student budget and is listed in the roadmap for honesty, not costed here.

9.2 Risk register

ID	Risk	Class	Mitigation / disposition
R-01	Convection exponent 0.8 is an unvalidated team choice; thermal margins may be off	ASSUMPTION	E1 calibrates it; model is conservative vs textbook range; margins carry the uncertainty
R-02	Paschen-minimum confusion: code gives 305 V, literature quotes ~327 V	COMPUTED vs LITERATURE	Both documented in §7.2; screening uses the code curve consistently; reviewer can reproduce
R-03	Hand-pump desiccator rig cannot reach or combine qualification conditions	PROPOSED	Honestly framed as partial pressure-response only; no qualification claimed from it
R-04	18650 cold-test safety (plating, mechanical stress)	PROPOSED	Branded cells, BMS, charge-lock < 0 °C, supervised tests only
R-05	Scope creep into HV arcing experiments	Safety rule	E2 is simulation-only, permanently; no HV supply in BOM
R-06	No accredited chamber access → no combined-environment qualification	FUTURE	Stated openly; simulator is pre-qualification screening, not a substitute
R-07	Misattributing electronics qualification to DIHAR	Reputational	Correction in §5/§10.4; DIHAR cited only for its actual domain
R-08	Price drift before purchase	ESTIMATE	Re-verify all prices on order day; total ≈ ■6,800 is planning only
R-09	Deployed app and simulation/ use different model implementations (battery lookup-table vs linear; Theta-JA vs lumped thermal); clearance-margin formula not in repo	COMPUTED	Both implementations disclosed in §5/§7.2 and never mixed; standing action: document the app formula and align or version the models
R-10	Site's 100 Wh mission pack is a scenario placeholder, not the 2S1P 22.2 Wh hardware	ASSUMPTION	§5 states the distinction; placeholder adjustable via pack_Wh, hardware sized at 22.2 Wh

VALIDATION ROADMAP · REFERENCES · LIMITATIONS

What happens in which order, what everything rests on, and what is not claimed

10.1 Phased validation roadmap

Phase	Content	Status
Phase 0 — now	Simulator + physics library; design screening; this blueprint	IMPLEMENTED
Phase 1	E1 thermal (freezer: cold at ambient pressure) + E3 cold discharge; TEST-PCB-01 fabricated	PROPOSED
Phase 2	Desiccator partial pressure-response (room temperature); leak-down observations	PROPOSED
Phase 3	Twin-board calibration loop; model recalibration; validation report v1	FUTURE VALIDATION
Phase 4	Accredited chamber, combined low-pressure + temperature (JSS 55555 Test No. 3 class)	FUTURE — outside student budget

What a freezer proves and does not prove: a freezer tests cold response at local ambient pressure — it does not reproduce high-altitude low pressure. A vacuum desiccator tests partial pressure response at room temperature — it does not reproduce coupled altitude conditions. Only a suitable environmental chamber applies controlled combined low pressure and temperature.

10.2 References

- [1] PIB, 3 Feb 2017 — Siachen winter clothing designed for below -50°C . pib.gov.in/PressReleasePage.aspx?PRID=1481715 [LITERATURE]
- [2] PIB, 10 Dec 2021 — troops use avalanche victim detectors/trackers; DGRE operates 39 observatories and automated weather stations. [LITERATURE]
- [3] Economic Times, 2015 — DRDO inaugurates Chang La station (17,600 ft, -40°C); secondary reporting. [LITERATURE, secondary]
- [4] Hach India — hand-operated vacuum pump, 63.5 cm (25 in) Hg vacuum (catalogue spec). [LITERATURE, catalogue]
- [5] IEC 60664-1 Table A.2 — altitude correction factors ($\times 1.14$ @ 3 km, $\times 1.29$ @ 4 km, $\times 1.48$ @ 5 km, $\times 1.70$ @ 6 km). Secondary citation; standard paywalled, not primary-verified. [LITERATURE, secondary]
- [6] IPC-2221B — B2/B3 altitude columns for clearance (not multiplicative factors). [LITERATURE]
- [7] JSS 55555 Rev. 3 (unofficial copy; controlled document) — Test No. 3 (Altitude): simultaneous low pressure + high/low temperature. [LITERATURE, unofficial]
- [8] MIL-STD-810H — Method 500.6 (Low Pressure/Altitude), Method 502.7 (Low Temperature). [LITERATURE]
- [9] U.S. Standard Atmosphere 1976, troposphere branch — as implemented in simulation/common/physics.py. [LITERATURE]
- [10] 18650 cell datasheet — NOT YET SELECTED. No cell datasheet is on file; the -20°C anchor is ASSUMPTION. [ASSUMPTION — CELL TO BE VERIFIED BEFORE PURCHASE]
- [11] physics.py + computation log (re-run 30 Sep 2026) — source of all COMPUTED values. github.com/jaisogani-ai/SIH2026 [COMPUTED]
- [12] HIMKAVACH simulator — himkavach.grok.me (screenshot §5, captured 30 Sep 2026). (IMPLEMENTED DIGITAL PROTOTYPE — software only; app source not in repo — see limitations)

10.3 Limitations & red-team disposition

- No verifiable public incident of electronics failing specifically from altitude/low-pressure/cold was found (search 30 Sep 2026) — none is claimed.
- DIHAR correction (see §5): DIHAR Leh is an agro-animal / life-sciences laboratory, not an electronics-hardening laboratory.
- Repo and app Paschen constants both give 305.3 V at the minimum (same B/A ratio); the ≈ 327 V textbook figure depends on the (A, B, γ) set and is not our computed result — disclosed, not hidden.
- IEC factors are cited from secondary engineering references; the standard itself was not primary-verified (paywalled).
- A vacuum desiccator is a partial, room-temperature pressure-response experiment — the earlier “Chang La in a jar” phrasing is retired.
- The deployed app’s battery lookup-table ‘literature’ anchors (77.4% @ -20°C , 88% @ -10°C , 32.8% @ -30°C) are implemented in the app JS; their source documents are not in the repo — REFERENCE TO VERIFY before citing as literature.
- Web-app source (himkavach.grok.me) is not in the GitHub repo; simulation/ is the reference implementation for this blueprint’s §7.2 anchors. Publishing the app source is a standing action.
- Reviewer verdict this document aims for: the hardware is not built, but the validation pathway is fully engineered and the next measurements are clear.